



Data Management

Week 4 - Gephi

Prof. Dr. Mehrdad Jalali



Introduction to Gephi



What is Gephi?

- Gephi is an **open-source network visualization tool** for exploring and analyzing complex graphs.
- Used in **social network analysis, scientific research, fraud detection, marketing, and more.**

Why Gephi?

- ✓ Easy-to-use graphical interface
- ✓ Supports **large-scale network visualization**
- ✓ Provides **statistical analysis** & community detection
- ✓ Exports graphs in various formats for reports & research

Installing Gephi

- Go to [Gephi.org](https://gephi.org) and download the latest version.
- Install **Java Runtime Environment (JRE)** if not already installed.
- Open Gephi after installation and explore the interface.

Understanding the Gephi Interface

- ❑ **Graph Window:** Displays network visualization
- ❑ **Data Laboratory:** Shows nodes and edges in tabular format
- ❑ **Statistics Panel:** Computes centrality, clustering, etc.
- ❑ **Layout Panel:** Applies different graph layouts
- ❑ **Filters Panel:** Helps to focus on specific parts of the network

Creating a Graph Manually in Gephi

Step 1: Open Gephi and Create a New Project

- Click **File** → **New Project**
- Select **Graph Window**

Step 2: Add Nodes (Entities)

- Go to **Data Laboratory** → **Nodes Table** → **Add Node**
- Name the nodes (e.g., Alice, Bob, Charlie)

Step 3: Add Edges (Connections)

- Switch to **Edges Table** → **Add Edge**
- Connect nodes with relationships (e.g., Alice → Bob)

Importing External Datasets into Gephi

■ Step 1: Download or Prepare a Graph File

- Supported formats: **CSV, GML, GraphML, Edge Lists**
- Example: Import a **Facebook friend network** (CSV format)

■ Step 2: Load the Data

- Go to **File** → **Open** and select the dataset
- Configure node/edge attributes and click **Next**

■ Step 3: Adjust Data Settings

- Define whether edges are **directed or undirected**
- Click **Finish** to generate the network

Applying Layouts for Better Visualization

■ Popular Layouts in Gephi:

- **Force Atlas:** Best for small-medium social networks
- **Fruchterman Reingold:** Evenly distributes nodes
- **Yifan Hu:** Good for large graphs

■ Step 1: Select a Layout

- Go to **Layout Panel** → Choose a layout (e.g., Force Atlas)
- Click **Run** and adjust parameters for better clarity

■ Step 2: Customize Node & Edge Appearance

- Use the **Appearance Panel** to change node size based on degree
- Adjust edge thickness and color

Computing Graph Metrics in Gephi

■ Step 1: Open the Statistics Panel

- Click **Statistics** (top-right corner)
- Run metrics like **Degree Centrality**, **Modularity**, **Clustering Coefficient**

■ Step 2: Interpret Results

- **Degree Centrality**: Who has the most connections?
- **Modularity**: Identifies communities
- **Betweenness Centrality**: Identifies key intermediaries

Filtering

■ Open the Filters Panel

- Use **Degree Filter** to show only highly connected nodes
- Apply **Giant Component Filter** to remove isolated nodes

Exporting Graphs for Presentation

■ Step 1: Export as an Image or PDF

- Go to **File** → **Export** → **SVG/PDF/PNG**
- Adjust resolution for publication

■ Step 2: Export as an Interactive Graph

- Use **Gephi Plugins** to export as an HTML interactive graph

Hands-on Exercise

1. **Manually create a small network** with 6 nodes and 9 edges.
2. **Import a sample dataset** (Facebook friends or co-authorship network).
3. **Apply a layout and customize the graph.**
4. **Compute network metrics** and interpret results.
5. **Export and submit a visual graph representation.**